POWER PULSE

Instrument Transformers Important Formulae

Notes by Balaji Sir

1 Ratios

1. $$K_n = \frac{\text{rated primary}}{\text{rated secondary}}, \qquad R = \frac{\text{actual primary}}{\text{actual secondary}}$$

K_n = nominal ratio; R = actual ratio

2. $$\%\text{ratio error} = \frac{K_n - R}{R} \times 100, \qquad \text{RCF} = \frac{R}{K_n}$$

RCF = ratio correction factor

3. $$\text{Turns ratio } n = \frac{N_s}{N_p}, \qquad \text{CT: } K_n \approx \frac{N_s}{N_p}, \quad \text{PT: } K_n \approx \frac{N_p}{N_s}$$

the actual ratio differs from the turns ratio because of the excitation current

2 Burden

4. $$\text{Burden (VA)} = I_s^2 \, Z_b \ \text{(CT)}, \qquad \text{Burden} = \frac{V_s^2}{Z_b} \ \text{(PT)}$$

the error increases with burden

3 Standard ratings and rules

5. $$\text{CT secondary: 1 A or 5 A;} \quad \text{PT secondary: 110 V (or } 110/\sqrt{3}\text{)}$$

standard secondary values

6. CT secondary must never be open-circuited (flux saturates, dangerous high voltage)

short the secondary before disconnecting the burden

7. PT secondary must never be short-circuited (excessive current)

4 Protection CT

8. $$\text{5P20: composite error } \leq 5\% \text{ at } 20 \times \text{rated current (accuracy limit factor 20)}$$

P = protection class